Experiment No. 8

Experiment Title	To implement Association Rule Mining (Apriori Algorithm) using RapidMiner/Weka tool
Course Outcomes	CO4: Implement the appropriate data mining methods like classification, clustering or Frequent Pattern mining on large data sets. CO5: Define and apply metrics to measure the performance of various data mining algorithms.
Date of Preparation	
Date of Submission	
Signature & Grade	

1. Aim

To implement Association Rule Mining on airline passenger data using the RapidMiner tool, to find frequent itemsets and association rules, and to read the rules using support and confidence.

2. Theory

2.1 Association Rule Mining

Association rule mining finds items that occur together in many transactions. A rule has the form IF antecedent THEN consequent, for example IF Type of Travel = Business travel THEN Class = Business. Here a transaction is one passenger and an item is one attribute value of that passenger.

2.2 Apriori, FP-Growth and RapidMiner

The Apriori algorithm first finds all frequent itemsets, which are combinations of items whose support is at least the minimum support. It uses the Apriori property: every subset of a frequent itemset is also frequent, so any combination that contains an infrequent item is dropped. It then builds the rules from the frequent itemsets and keeps those whose confidence is at least the minimum confidence.

RapidMiner does not have an operator named Apriori. It uses the FP-Growth operator to find the frequent itemsets and the Create Association Rules operator to make the rules. FP-Growth finds the same frequent itemsets as Apriori, and therefore the same rules; it only uses a faster method, which stores the data in a compact tree instead of generating and testing many candidate itemsets.

2.3 Measures

Measure	Formula	Meaning
Support	support(A, B) = transactions with A and B / all transactions	How common the rule is: the share of passengers that have all items of the rule.
Confidence	confidence(A → B) = support(A, B) / support(A)	Among passengers that have A, the share that also have B.
Lift	lift(A → B) = confidence(A → B) / support(B)	How much more often B occurs with A than by chance. Lift above 1 means a positive association.

3. Dataset and Tool

Dataset: Airline Passenger Satisfaction (Kaggle), file train.csv, with 103,904 passengers.

Source: kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction

Tool: RapidMiner (Altair AI Studio, Educational license).

Association rules need categories, not numbers. The data was therefore prepared so that every attribute value became an item (a true/false column) before the rules were mined.

4. Procedure

1. Retrieve: the dataset was loaded from the Local Repository.

2. Select Attributes: only the columns needed for the rules were kept, such as Customer Type, Type of Travel, Class, Age, Flight Distance, the service ratings and satisfaction.
3. Discretize: the numeric columns Age and Flight Distance were split into 3 ranges, for example Age = range2 [33 - 59] and Flight Distance = range1 [-∞ - 1681.667].
4. Numerical to Polynominal: the service ratings were turned into categories, so each rating value is an item, for example Online boarding = 4.
5. Nominal to Binominal: every category value became a true/false column (an item).
6. FP-Growth: finds the frequent itemsets for the chosen minimum support.
7. Create Association Rules: builds the rules from the frequent itemsets, using confidence as the criterion with the chosen minimum confidence.
8. The rules and the frequent itemsets were connected to the result ports and the process was run.5. Implementation and Output

5.1 The RapidMiner Process

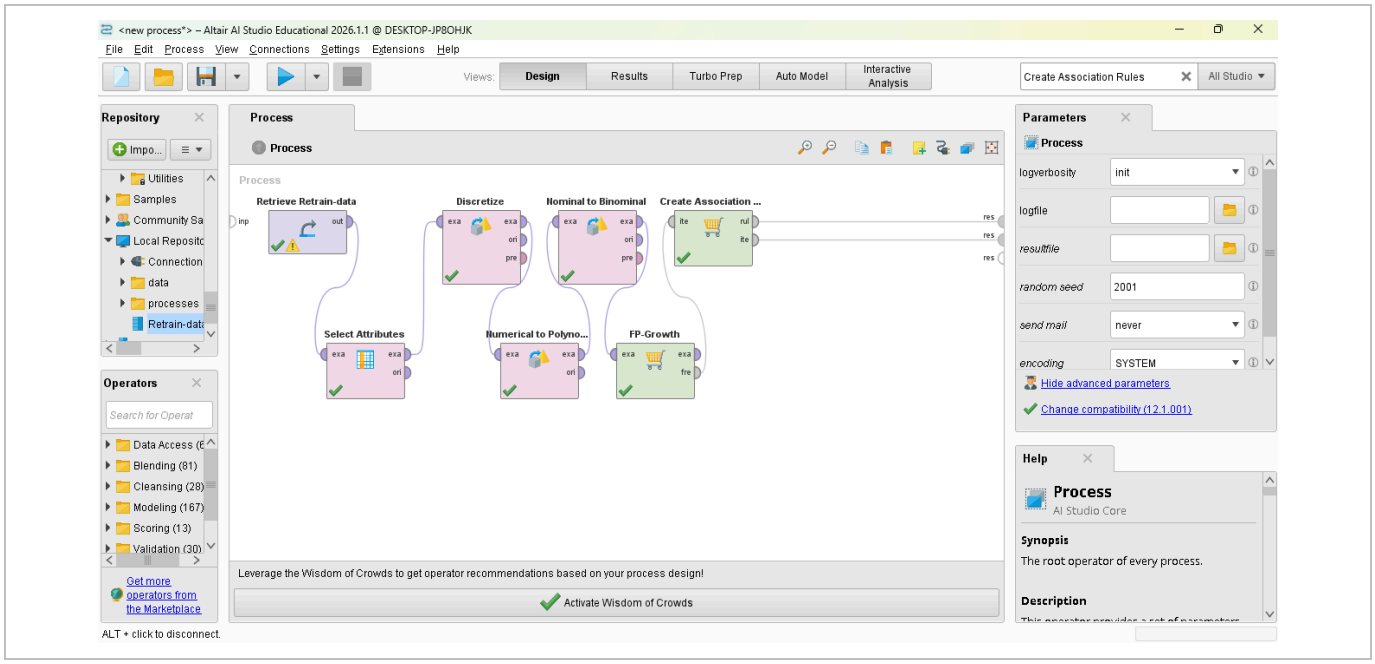


Figure 1: The association rule process in RapidMiner

The process runs from Retrieve through Select Attributes, Discretize, Numerical to Polynominal and Nominal to Binominal to FP-Growth and Create Association Rules. All operators show a green check. The rules (rul) and the item sets (ite) of Create Association Rules are connected to the result ports.

5.2 Association Rules: Graph View

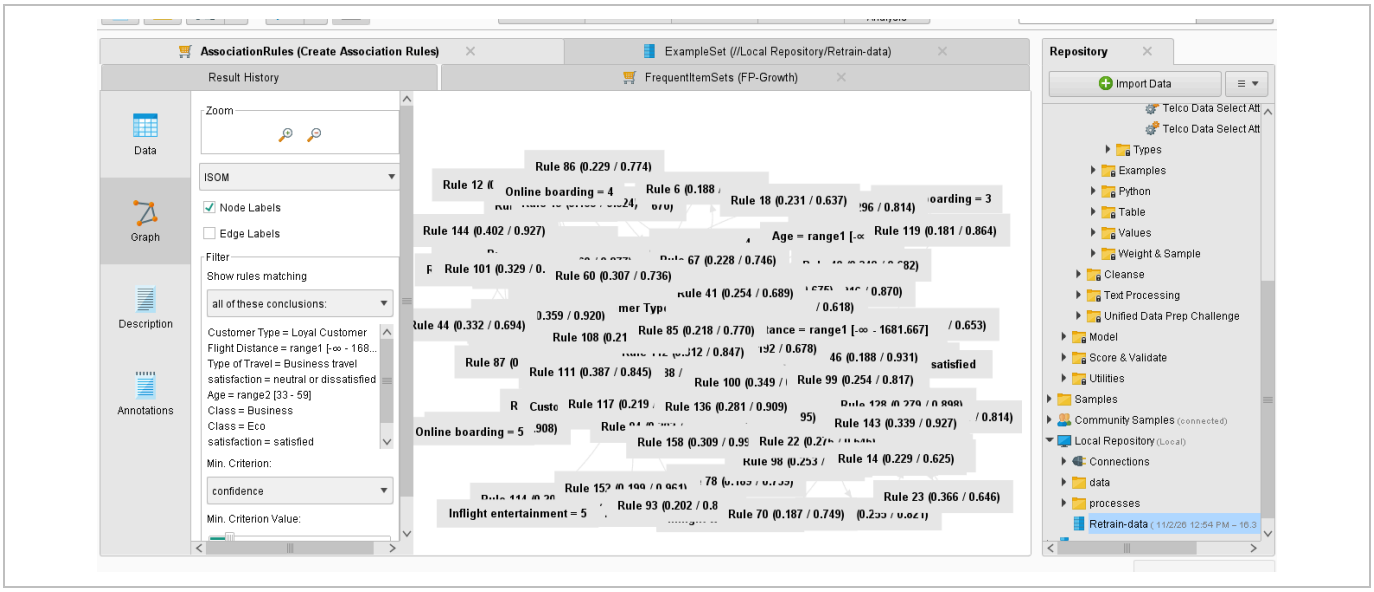


Figure 2: Association rules in the graph view

The graph view draws the items and the rules as nodes. Each rule node is labelled Rule N (support / confidence); for example Rule 144 has a support of 0.402 and a confidence of 0.927, and Rule 143 has a support of 0.339 and a confidence of 0.927. The rule numbers go up to at least 158, so at least 158 rules were found. The supports shown in the graph are between about 0.18 and 0.40, and the confidences are between about 0.62 and 0.99. The filter on the left lists the items that can be chosen as the conclusion, for example satisfaction = satisfied and satisfaction = neutral or dissatisfied, and the minimum criterion slider (confidence) hides weaker rules. Because there are many rules the graph is crowded, so the table view is used to read the rules.

5.3 Association Rules: Table View

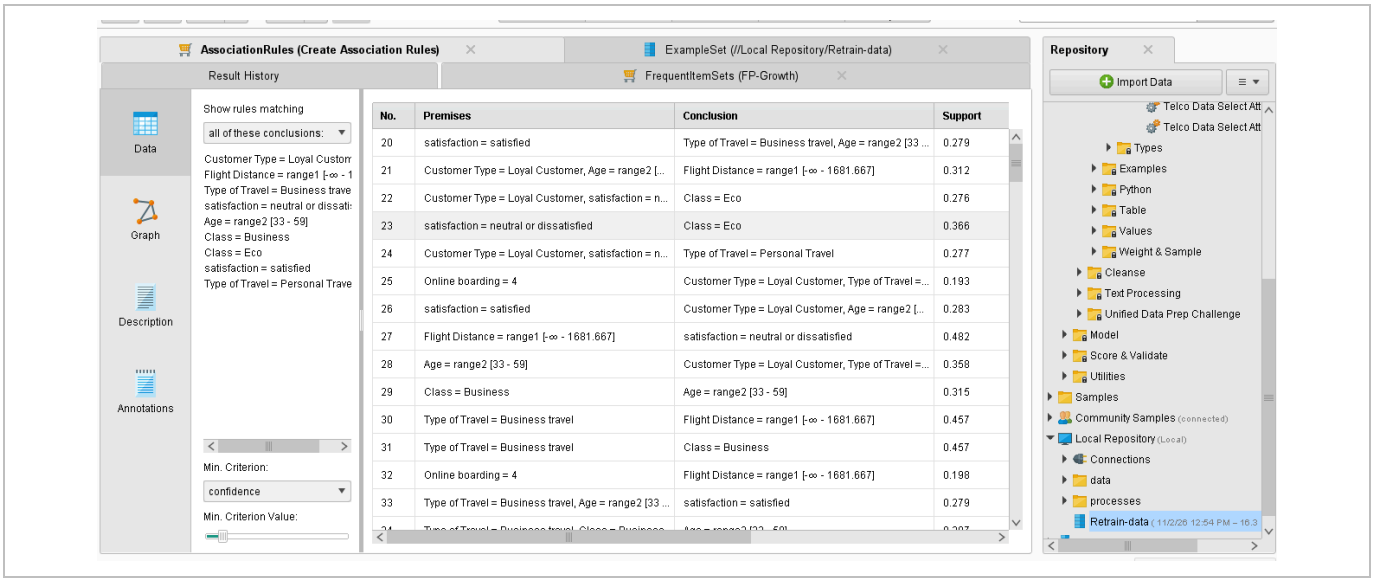


Figure 3: Association rules in the table view (premises, conclusion and support)

The table lists each rule with its premises, its conclusion and its support. Some of the rules shown are:

Rule	Premises	Conclusion	Support
27	Flight Distance = range1 [-∞ - 1681.667]	satisfaction = neutral or dissatisfied	0.482
30, 31	Type of Travel = Business travel (rule 30); Class = Business (rule 31)	Class = Business (rule 30); Type of Travel = Business travel (rule 31)	0.457
23	satisfaction = neutral or dissatisfied	Class = Eco	0.366
28	Age = range2 [33 - 59]	Customer Type = Loyal Customer, Type of Travel = …	0.358
33	Type of Travel = Business travel, Age = range2 [33 - 59]	satisfaction = satisfied	0.279

Reading the rows: 48.2% of the passengers are on a short flight (up to 1,681.667) and are neutral or dissatisfied. 45.7% of the passengers travel for business in Business class, and rules 30 and 31 are the same itemset written in both directions, so they have the same support. 27.9% of the passengers are business travellers aged 33 to 59 who are satisfied.

Checking a measure: Rule 23 (neutral or dissatisfied → Class = Eco) has a support of 0.366 in the table and a confidence of 0.646 in the graph label. This agrees with the formula, because 0.366 / 0.567 = 0.646, where 0.567 is the share of neutral or dissatisfied passengers in the dataset (56.7%). So 64.6% of the neutral or dissatisfied passengers travel in Eco class.

6. Conclusion

Association rule mining was performed in RapidMiner on the Airline Passenger Satisfaction data, using FP-Growth to find the frequent itemsets and Create Association Rules to build the rules, after the columns had been turned into true/false items. At least 158 rules were found, with supports between about 0.18 and 0.48. The rules show that business travel and Business class occur together in 45.7% of the passengers, that short flights go with neutral or dissatisfied passengers (support 0.482), that 64.6% of the neutral or dissatisfied passengers travel in Eco class, and that business travellers aged 33 to 59 are often satisfied (support 0.279).